

Lecture 9 :

SVD in Machine Learning

Ridge Regression

$(x_i, y_i) \in \mathbb{R}^p \times \mathbb{R}$, $i=1, \dots, n$

until now, we considered $n > p$, $X = \begin{bmatrix} -x_1^T- \\ -x_2^T- \\ \vdots \\ -x_n^T- \end{bmatrix}$ w/ p LI columns. Then $\hat{w}_{LS} = (X^T X)^{-1} X^T y$

These assumptions were necessary to ensure there was a unique set of weights minimizing squared error — and that $X^T X$ had an inverse.

$$\text{Let } X = U \Sigma V^T \Rightarrow X^T X = (U \Sigma V^T)^T (U \Sigma V^T) = V \Sigma^T U^T U \Sigma V^T = V \Sigma^T \Sigma V^T$$

$$\text{Then } (X^T X)^{-1} = (V \Sigma^T \Sigma V^T)^{-1} = (V^T)^{-1} (\Sigma^T \Sigma)^{-1} V^{-1} = V (\Sigma^T \Sigma)^{-1} V^T$$

Now $\Sigma^T \Sigma = \begin{bmatrix} \sigma_1^2 & & & \\ & \sigma_2^2 & & \\ & & \ddots & \\ & & & \sigma_r^2 \\ & & & & 0 \\ & & & & & \ddots \\ & & & & & & 0 \end{bmatrix}$, so we can only invert $\Sigma^T \Sigma$ if all the σ_i 's > 0

That is,

X has p LI columns $\Leftrightarrow X^T X$ invertible $\Leftrightarrow X$ is positive definite $\Leftrightarrow$ all singular values > 0

If X has $r < p$ LI columns, then $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0 = \sigma_{r+1} = \sigma_{r+2} = \dots = \sigma_p$

Can we still learn a predictor when X has $r < p$ LI columns?

e.g., if $n < p$ (more features than training samples), then there are at most n LICs

Recall ridge regression:

$$\hat{\underline{w}}_{\lambda} = \underset{\underline{w}}{\operatorname{argmin}} \quad \|\underline{y} - X\underline{w}\|_2^2 + \lambda \|\underline{w}\|_2^2 = (X^T X + \lambda I)^{-1} X^T \underline{y}$$

Let $X = U\Sigma V^T$, as before.

$$\begin{aligned} \text{Then } X^T X + \lambda I &= V \Sigma^T \Sigma V^T + \lambda I = V \Sigma^T \Sigma V^T + \lambda \underbrace{V V^T}_I = V \Sigma^T \Sigma V^T + V(\lambda I) V^T \\ &= V(\Sigma^T \Sigma + \lambda I) V^T \end{aligned}$$

$$\Sigma^T \Sigma = \begin{bmatrix} \sigma_1^2 & & & \\ & \sigma_2^2 & & \\ & & \ddots & \\ & & & \sigma_r^2 \\ & & & & 0 \end{bmatrix} \Rightarrow \Sigma^T \Sigma + \lambda I = \begin{bmatrix} \sigma_1^2 + \lambda & & & \\ & \sigma_2^2 + \lambda & & \\ & & \ddots & \\ & & & \sigma_r^2 + \lambda \\ & & & & 0 \end{bmatrix}$$

Even if $\sigma_p = 0$,
 $\sigma_p^2 + \lambda > 0$

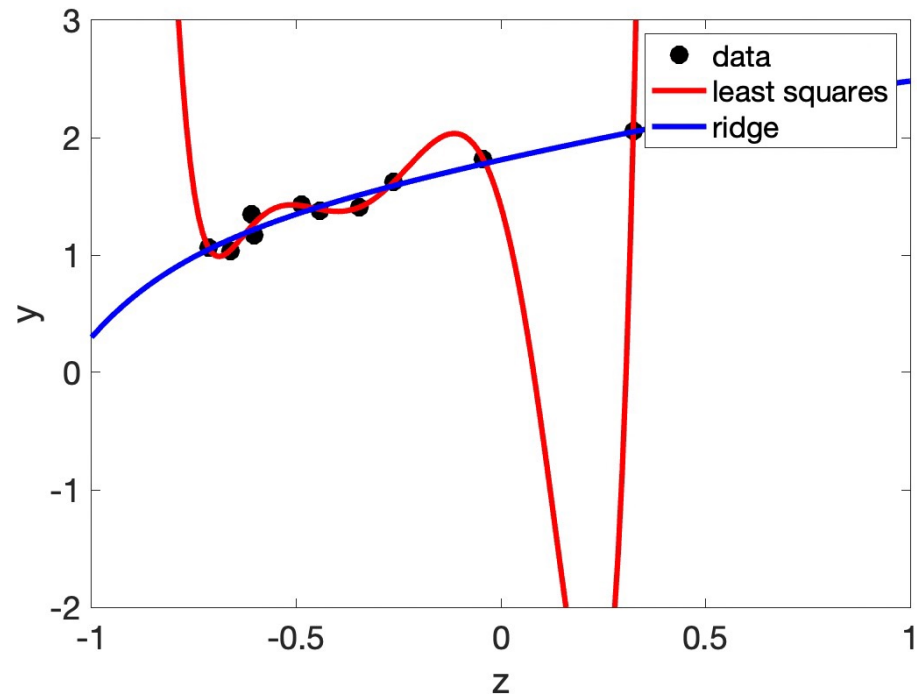
$\Rightarrow \Sigma^T \Sigma$ may not be invertible
but $\Sigma^T \Sigma + \lambda I$ is always
invertible

$\Rightarrow$ We can always compute the ridge regression estimate,
even when a unique least squares estimate does
not exist

$$\begin{aligned}\hat{\underline{w}}_{\lambda} &= (X^T X + \lambda I)^{-1} X^T y \\ &= V(\Sigma^T \Sigma + \lambda I) V^T V \Sigma^T U^T y \\ &= V(\Sigma^T \Sigma + \lambda I)^{-1} \Sigma^T U^T y\end{aligned}$$

$$\begin{bmatrix} \frac{\sigma_1}{\sigma_1^2 + \lambda} & & & \\ & \frac{\sigma_2}{\sigma_2^2 + \lambda} & & \\ & & \ddots & \\ & & & \frac{\sigma_p}{\sigma_p^2 + \lambda} \end{bmatrix}$$

Ridge regression can help prevent overfitting



WHY?

Imagine X has some very small singular values, e.g. 10^{-6} .

Also, imagine $y = X \underline{w}^* + \underline{\varepsilon}$ where $\underline{\varepsilon}$ is random error or noise added to each sample

Then
$$\hat{\underline{w}}_{LS} = (X^T X)^{-1} X^T y = (X^T X)^{-1} X^T (X \underline{w}^* + \underline{\varepsilon})$$

$$= \cancel{(X^T X)^{-1} X^T X} \underline{w}^* + (X^T X)^{-1} X^T \underline{\varepsilon}$$

$$= \underbrace{\underline{w}^*}_{\substack{\text{good!} \\ \text{what we want}}} + \underbrace{V^T (\Sigma^T \Sigma)^{-1} \Sigma^T U^T}_{\begin{bmatrix} 1/\sigma_1 & & \\ & 1/\sigma_2 & \\ & & \ddots \\ & & & 1/\sigma_p \end{bmatrix}} \underline{\varepsilon}$$

$\sim 10^6 \Rightarrow$ a little noise in $\underline{\varepsilon}$ gets multiplied by a huge number $\Rightarrow$ least squares might fit observations very closely but give strange predictions on test data

In contrast,
$$\hat{\underline{w}}_R = (X^T X + \lambda I)^{-1} X^T (X \underline{w}^* + \underline{\varepsilon})$$

$$= (X^T X + \lambda I)^{-1} X^T X \underline{w}^*$$

$$+ (X^T X + \lambda I)^{-1} X^T \underline{\varepsilon}$$

$$= V (\Sigma^T \Sigma + \lambda I)^{-1} \Sigma^T \Sigma V^T \underline{w}^*$$

$$+ V (\Sigma^T \Sigma + \lambda I)^{-1} \Sigma^T U^T \underline{\varepsilon}$$

$$\begin{bmatrix} \frac{\sigma_1^2}{\sigma_1^2 + \lambda} & & \\ & \ddots & \\ & & \frac{\sigma_p^2}{\sigma_p^2 + \lambda} \end{bmatrix}$$

$$\frac{\sigma_i}{\sigma_i^2 + \lambda} \approx \begin{cases} 1/\sigma_i & \sigma_i \gg \lambda \\ 1/\lambda & \sigma_i \approx 0 \end{cases}$$

$\approx \underline{w}^*$
for small λ

none of these values "blow up" by being close to dividing by 0, so we do not magnify noise $\Rightarrow$ better predictions on test data

The Elephant In the Room - how do we choose λ ?

1. Split data into 2 sets: $(\underline{x}_i, y_i), i=1, \dots, m$ = training set
 $(\underline{x}_i, y_i), i=m+1, \dots, n$ = validation set.

2 For each $\lambda \in \{\lambda_1, \lambda_2, \dots, \lambda_p\}$, find $\hat{\underline{w}}_\lambda$ using training data

Measure the loss of each λ using validation set: $L_\lambda = \sum_{i=m+1}^n (y_i - \underline{x}_i^\top \hat{\underline{w}}_\lambda)^2$

choose λ with smallest L_λ .

Alternative to ridge regression (when some singular values = 0 so many $\underline{w}$ fit $\underline{y}$ perfectly)

with least squares, we needed to compute $(\Sigma^\top \Sigma)^{-1} \Sigma^\top = \begin{bmatrix} 1/\sigma_1 & & & \\ & 1/\sigma_2 & & \\ & & \ddots & \\ & & & 1/\sigma_p \end{bmatrix}$,
which was problematic for any $\sigma_i = 0$

Alternative: define pseudoinverse of Σ as

$$(\Sigma^+)_{ii} = \begin{cases} 1/\sigma_i & \text{if } \sigma_i > 0 \\ 0 & \text{otherwise} \end{cases}$$

that is, Σ^+ corresponds to taking the transpose of Σ and only inverting the nonzero diagonal entries

Special case: $p > n$, X has n linearly independent rows.

$$\Rightarrow \Sigma \sim n \times p$$

$$\boxed{\begin{matrix} X \\ n \times p \end{matrix}} = \boxed{\begin{matrix} U \\ n \times n \end{matrix}} \boxed{\begin{matrix} \Sigma \\ n \times p \end{matrix}} \boxed{\begin{matrix} V^T \\ p \times p \end{matrix}}$$

$$\Sigma = \left[\begin{array}{ccc|c} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \\ & & & \text{---} \end{array} \right] \Rightarrow \Sigma^+ = \left[\begin{array}{ccc|c} 1/\sigma_1 & & & \\ & 1/\sigma_2 & & \\ & & \ddots & \\ & & & 1/\sigma_n \\ \hline & & & \text{---} \end{array} \right] \left. \begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \end{array} \right\} \begin{array}{l} n \text{ nonzero} \\ \text{singular vals.} \end{array} \quad \left. \begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \\ \text{ } \end{array} \right\} \begin{array}{l} p-n \text{ columns} \\ \text{of zeros.} \end{array}$$

$$= \Sigma^T (\Sigma \Sigma^T)^{-1}$$

pseudoinverse solution $\hat{\underline{w}} = V \Sigma^+ U^T \underline{y}$

$$\begin{aligned} &= V \Sigma^T (\Sigma \Sigma^T)^{-1} U^T \underline{y} \\ &= X^T (X X^T)^{-1} \underline{y} \end{aligned}$$

Claim: when X has n LI rows, the choice $\hat{w} = V \Sigma^+ U^T y = X^T (X X^T)^{-1} y$ has the smallest norm $\|w\|_2^2$ of any w satisfying $y = Xw$.

Proof: for any w ,

$$\|w\|_2^2 = \|w - \hat{w} + \hat{w}\|_2^2 = \|w - \hat{w}\|_2^2 + 2(w - \hat{w})^T \hat{w} + \|\hat{w}\|_2^2$$

Assume $Xw = y = X\hat{w} \Rightarrow \underline{X(w - \hat{w}) = 0}$.

$$\begin{aligned} \text{Then } (w - \hat{w})^T \hat{w} &= (w - \hat{w})^T X^T (X X^T)^{-1} y \\ &= (\underline{X(w - \hat{w})})^T (X X^T)^{-1} y \\ &= \underline{0} \end{aligned}$$

$$\Rightarrow \|w\|_2^2 = \|w - \hat{w}\|_2^2 + \|\hat{w}\|_2^2 \geq \|\hat{w}\|_2^2 \Rightarrow \hat{w} \text{ has smallest norm!}$$

Why does minimum norm make sense?

$$\text{e.g. } X = \begin{bmatrix} 1 & 0 & 0.1 \\ 0 & 1 & 0 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

$$\text{possible } \underline{w}'\text{s include } \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix}, \begin{bmatrix} -10 \\ 0 \\ 100 \end{bmatrix}, \begin{bmatrix} -1000 \\ 0 \\ 10000 \end{bmatrix}$$

$\uparrow$ minimum norm!